

AVINASH MADASU

avinashmadasu17@gmail.com ♦ [Github](#) ♦ [LinkedIn](#) ♦ [Google Scholar](#) ♦ [Semantic Scholar](#) ♦ [Website](#)

EDUCATION

Master of Science, Computer Science
University of North Carolina at Chapel Hill

August 2021 - May 2023

Bachelor of Technology, Computer Science
National Institute of Technology Tiruchirappalli

July 2014 - May 2018

EXPERIENCE

Intel Corporation
AI Research Scientist

June 2023 - present

- Architected an agentic long-context inference system that recursively plans, decomposes, and delegates 100K+ token workloads into bounded sub-calls, sustaining 43% accuracy on LongBench across various context lengths while handling inputs beyond practical single-prompt limits.
- Designed a token-aware heterogeneous router for RLM agents, dynamically assigning sub-calls across local small models and frontier-model backends; offloaded 100% of eligible leaf calls to Xeon while preserving all-cloud accuracy, enabling backend-agnostic execution from cloud to fully local inference.
- Launched and led cross-organizational efforts to scale an enterprise RAG platform on Xeon infrastructure, owning stakeholder alignment, cross-team execution, and a 12-month roadmap to improve retrieval quality, expand evaluation, and enable production deployment for external customers.
- Pioneered a novel multimodal Chain-of-Thought (CoT) reasoning approach, leveraging synthetic data generation to substantially improve performance in reasoning models like LLaVA.
- Built a large-scale diffusion-model pipeline generating 10M+ counterfactual images across distributed accelerators, distilled into a 171K-pair benchmark to systematically audit and mitigate intersectional bias in production vision-language models.

Intel Corporation
Research intern

May 2022 - Aug 2022

- Architected a scalable pseudo-labeling and knowledge-transfer pipeline expanding video search to 50+ languages, eliminating large-scale manual labeling and enabling zero-shot multilingual retrieval across internal platforms.
- Published in ECIR 2023, a top Information Retrieval conference, and won the Best Student Paper Award and also the first work to leverage the Gaudi AI accelerator.

UNC Chapel Hill
Graduate Research Assistant

August 2021 - May 2023

Advisor: Prof. Gedas Bertasius

- Proposed a new interactive video retrieval system that leverages dialog with the users. The proposed approach requires minimum rounds of dialog to outperform the human dialog.
- Proposed interactive system outperformed static video retrieval models by 2.8 points in R@1 and 10 points in R@5. This system is the first to use generative models (BART, T5) to perform open ended dialog with the users.

UNC Chapel Hill
Graduate Research Assistant

August 2021 - November 2021

Advisor: Prof. Shashank Srivastava

- Explored the inductive biases in pre-trained language models for solving non-linguistic reasoning tasks.

- Proposed a set of 19 diverse non-linguistic tasks involving quantitative computations, recognizing regular expressions and reasoning over strings and published this work in EMNLP 2022.

Samsung R&D Institute India - Bangalore

June 2018 - July 2021

Senior Software Engineer

- Built and deployed GDPR-compliant named entity recognition systems protecting phone numbers, personal identities, and financial data across billions of voice interactions in global environments.
- Led the design and deployment of a lightweight detection framework achieving 90% accuracy across 10 applications, reducing intent and application misidentification and improving end-to-end command success rates in Bixby voice assistant systems.
- Built and deployed a large-scale ULMFiT LLM based feedback prediction system achieving 90% failure-detection accuracy across billions of user text commands, enabling systematic identification and correction of Bixby interaction failures in production across Clock, Messages, Calendar, and additional core applications.

IIT Patna

July 2020 - January 2021

Research Assistant

Advisor: Prof. Asif Eqbal

- Proposed an End-to-End model capable of identifying slots from user utterances without external slot labels.
- Published one of the earliest works to use large scale pretraining in dialog systems which significantly improved slot identification accuracy and dialog generation.

National Institute of Technology, Tiruchirappalli

July 2017 - May 2018

Research Assistant

Advisor: Prof. Sivasankar

- Designed efficient lexicon based feature selection techniques that achieved excellent performance in resource scarce conditions. This work was published in the journal of multimedia tools and applications.
- Conducted a systematic study on the advantages and disadvantages of statistical feature selection techniques vs neural network techniques (word2vec, Doc2vec).
- Explored the possibility of combining text reviews with numerical ratings to design a hybrid recommender system.

PUBLICATIONS

1. [Is Your Paper Being Reviewed by an LLM? Benchmarking AI Text Detection in Peer Review](#)
Sungduk Yu, Man Luo, [Avinash Madasu](#), Vasudev Lal, Phillip Howard
ICLR 2026
2. [Learning from Reasoning Failures via Synthetic Data Generation](#)
Gabriela Ben Melech Stan, Estelle Aflalo, [Avinash Madasu](#), Vasudev Lal, Phillip Howard
AAAI 2026
3. [Pruning the Paradox: How CLIP's Most Informative Heads Enhance Performance While Amplifying Bias](#)
[Avinash Madasu](#), Vasudev Lal, Phillip Howard
EMNLP 2025 (Oral) (Senior Area Chair Recommendation)
4. [Affective Visual Dialog: A Large-Scale Benchmark for Emotional Reasoning Based on Visually Grounded Conversations](#)
Kilichbek Haydarov, Xiaoqian Shen, [Avinash Madasu](#), Mahmoud Salem, Jia Li, Gamaleldin Elsayed, Mohamed Elhoseiny
ECCV 2024

5. [Probing and Mitigating Intersectional Social Biases in Vision-Language Models with Counterfactual Examples](#)
Phillip Howard, [Avinash Madasu](#), Tiep Le, Gustavo Lujan Moreno, Anahita Bhiwandiwalla, Vasudev Lal
CVPR 2024
6. [ICSVR: Investigating Compositional and Syntactic Understanding in Video Retrieval Models](#)
[Avinash Madasu](#), Vasudev Lal
CVPR (MMFM workshop) 2024
7. [Analyzing Zero-Shot Abilities of Vision-Language Models on Video Understanding Tasks](#)
[Avinash Madasu](#), Anahita Bhiwandiwalla, Vasudev Lal
NeurIPS (R0-FoMo workshop) 2023
8. [Probing Intersectional Biases in Vision-Language Models with Counterfactual Examples](#)
Phillip Howard, [Avinash Madasu](#), Tiep Le, Gustavo Lujan Moreno, Vasudev Lal
NeurIPS (Diffusion models workshop) 2023
9. [MuMUR: Multilingual Multimodal Universal Retrieval](#)
[Avinash Madasu](#), Estelle Guez Aflalo, Gabriela Ben Melech Stan, Shachar Rosenman, Shao-Yen Tseng, Gedas Bertasius, Vasudev Lal
Information Retrieval Journal
10. [A Unified Framework for Slot based Response Generation in a Multimodal Dialogue System](#)
Mauajama Firdaus*, [Avinash Madasu*](#), Asif Eqbal
Journal of Multimedia Tools and Applications
11. [Is Multi-Modal Vision Supervision Beneficial to Language?](#)
[Avinash Madasu](#), Vasudev Lal
CVPR (NFVLR workshop) 2023
12. [A Unified Framework for Emotion Identification and Generation in Dialogues](#)
[Avinash Madasu*](#), Mauajama Firdaus*, Asif Eqbal
EACL (SRW workshop) 2023
13. [Improving video retrieval using multilingual knowledge transfer](#)
[Avinash Madasu](#), Estelle Guez Aflalo, Gabriela Ben Melech Stan, Shao-Yen Tseng, Gedas Bertasius, Vasudev Lal
ECIR 2023 (Best Student Paper Award)
14. [What do Large Language Models Learn beyond Language?](#)
[Avinash Madasu](#), Shashank Srivastava
EMNLP (Findings) 2022
15. [Learning to Retrieve Videos by Asking Questions](#)
[Avinash Madasu](#), Junier Oliva, Gedas Bertasius
ACM Multimedia 2022
16. [Sequential Domain Adaptation through Elastic Weight Consolidation for Sentiment Analysis](#)
[Avinash Madasu](#), Vijjini Anvesh Rao
ICPR 2020

17. [A Position Aware Decay Weighted Network for Aspect based Sentiment Analysis](#)
[Avinash Madasu](#), Vijjini Anvesh Rao
NLDB 2020
18. [Sequential Learning of Convolutional Features for Effective Text Classification](#)
[Avinash Madasu](#), Vijjini Anvesh Rao
EMNLP 2019
19. [Efficient Feature Selection techniques for Sentiment Analysis](#)
[Avinash Madasu](#), Sivasankar E
Journal of Multimedia Tools and Applications
20. [Gated Convolutional Neural Networks for Domain Adaptation](#)
[Avinash Madasu](#), Vijjini Anvesh Rao
NLDB 2019
21. [Effectiveness of Self Normalizing Neural Networks for Text Classification](#)
[Avinash Madasu](#), Vijjini Anvesh Rao
CICLing 2019
22. [A Study of Feature Extraction techniques for Sentiment Analysis](#)
[Avinash Madasu](#), Sivasankar E
IEMIS 2018

PREPRINTS

1. [Multimodal Dialogue Modeling: Simultaneous Intent Recognition and Response Generation](#)
[Avinash Madasu*](#), Mauajama Firdaus*, Asif Ekbali

VOLUNTEER EXPERIENCE

Conference Reviewer: CVPR 2026, ICCV 2025, EMNLP 2025, COLM 2024, CVPR 2024, ICLR 2024, AAAI 2024, EMNLP 2023, AMLC 2023, NeurIPS 2023, BMVC 2023, CoLLA 2023, ACL 2023, CVPR 2023, EACL 2023, ACL 2022, ACL 2021, ICON 2020

Journal Reviewer: Machine learning, TMLR, Computer Speech & Language

Workshop Reviewer: ICLR - MoFo 2023, ICLR - MRL 2023, SocialNLP

PRESS

- [The Jerusalem Post](#)
- [Outlook India](#)
- [Analytics India Magazine](#)

AWARDS

- [Outstanding reviewer - CVPR 2026](#)
- [Outstanding reviewer - ACL 2023](#)
- Best Student Paper Award - ECIR 2023.
- ACM grant to attend ACM Multimedia 2022 conference.
- Samsung Citizen Award 2019, 2020 (Research) (4/9000).

OPEN SOURCE DEEP LEARNING FRAMEWORKS CONTRIBUTIONS

- [gluonnlp](#) (Contributor and Member of Distributed Machine Learning Community- dmlc).
- [serve](#)
- [pytorch-optimizer](#).
- [catalyst](#)
- [AllenNLP](#)

RELEVANT LINKS

- Google Scholar: <https://scholar.google.com/citations?user=YRe0ruYAAAAJ&hl=en>
- DBLP: <https://dblp.org/pid/241/5153.html>
- Semantic Scholar: <https://www.semanticscholar.org/author/Avinash-Madasu/115098946>
- ORCID iD: <https://orcid.org/0000-0002-3802-7618>